$$= e\left[\frac{a}{2} - x\right] = a - ex \qquad \ldots (1)$$

and

$$SP' = ePM'$$
$$= e[CN + CZ']$$
$$= e\left[x + \frac{a}{e}\right] = ex + a \qquad \ldots (2)$$

Hence, $SP + S'P = a - ex + a + ex = 2a$

Remark

When $b = a$, the equation $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$, becomes $(x-h)^2 + (y-k)^2 = a^2$ the equation of circle with centre (h, k) and radius a.

When $b = a, e = \sqrt{1 - \frac{a^2}{a^2}} = 0$. **Hence the eccentricity of the circle is zero.**

Furthere, $\frac{SP}{PM} = 0$ implies $PM \to \infty$. That is, the directrix of the circle is at infinity.

Remark

Auxiliary circle or circumcircle is the circle with length of major axis as diameter and **Incircle** is the circle with length of minor axis as diameter. They are given by $x^2 + y^2 = a^2$ and $x^2 + y^2 = b^2$ respectively.

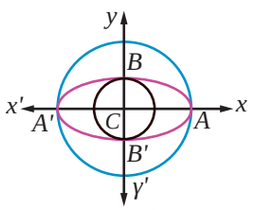


Fig.5.27

5.3.4 Hyperbola

We invoke that a hyperbola is the locus of a point which moves such that its distance from a fixed point (focus) bears a constant ratio (eccentricity) greater than unity its distance from its directrix, bearing a constant ratio e $(e > 1)$.

(i) Equation of a Hyperbola in standard form with centre at (0, 0)

Let S be a focus, l be the directrix line, e be the eccentricity $e > 1$ and $P(x, y)$ be the moving point. Draw SZ and PM perpendicular to l.

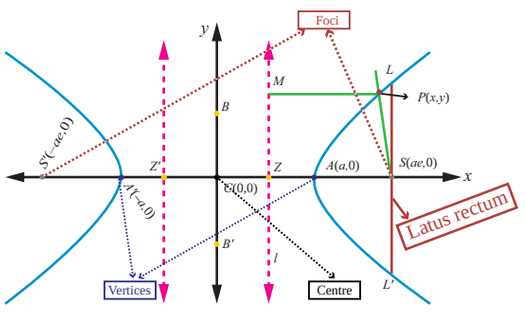


Fig.5.28

Let A and A' be the points which divide SZ internally and externally in the ratio $e:1$ respectively.

Let $AA' = 2a$. Let the point of intersection of the perpendicular bisector with AA' be C. Then CA = CA' = a Choose C as origin and the line CZ produced as x-axis and the perpendicular bisector of AA' as y-axis.

By definition, $\frac{AS}{AZ} = e$ and $\frac{A'S}{A'Z} = e$.

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